



UNDER CONSTRUCTION

Workers assemble Langley's Great Aerodrome on its launch track on top of a houseboat, ready for the first flight. An elaborate catapult mechanism had been created to propel the flying machine along the track and into the air.

Langley wrote, "the funds for continuing the work were exhausted, rendering it impossible to wait until spring for more suitable weather for making a test..." This time the attempt took place off Anacostia, at the edge of Washington, where crowds of onlookers lined the riverside. The brave Manly once more shot along the launch track and for a moment the huge machine lifted into the air, before the whole tail section sadly crumpled and broke away. Plunged into the icy water, Manly was trapped under the wreckage, but pulled himself free.

Stung by the ridicule heaped upon him, Langley blamed the failure on a faulty launch mechanism, insisting his Aerodrome could have flown.

As the 19th century drew to a close, the attaching of an engine to some form of glider had suddenly become more feasible through the development of the internal combustion engine,

which had the potential to generate more power per weight than any steam engine. Lilienthal's disciple Percy Pilcher was the first would-be aviator to develop a gasoline aero-engine. He intended to use the 4hp power plant to drive a propeller attached to one of his gliders. But in September 1899, while the powered machine was still being assembled, Pilcher was killed when his glider fell apart during a demonstration flight.

The deaths of Lilienthal and Pilcher were a major setback for those who believed in gliding as the route to powered manned flight. The road seemed open for followers of the power-focused tradition to triumph, in the person of distinguished American scientist Samuel Pierpont Langley. Dismissive of Lilienthal and his followers, Langley believed that the application of sufficient power to an aerodynamically stable machine would solve the problem of flight. In 1896, he felt he had proved his point by flying steam-powered model aircraft off the roof of a houseboat on the Potomac River. The models, which he called Aerodromes from the Greek for "air runners," had a wingspan of around 14ft (4.25m). One flew for 1 minute 30 seconds and another for 1 minute 45 seconds.

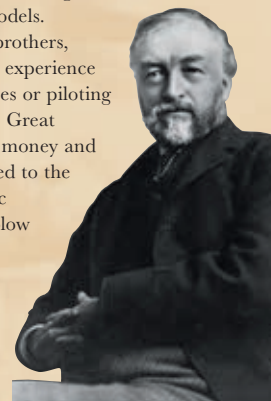
Had Langley ended his work there, his contribution to aviation history would have been a resounding success. But the temptation to pursue manned flight proved too strong. In 1898 the United States went to war with Spain. The US War Department offered Langley generous funding to produce a flying machine, regarded as a potential weapon of war. With a budget of \$50,000, plus the resources of the Smithsonian to call on, Langley fondly expected to achieve manned flight by the end of 1899. Yet delay followed delay. Langley settled on a gasoline engine to power his airplane, but it took years to develop one with the power-to-weight ratio he required. Building a full-scale version of his Aerodrome models also proved taxing for the Smithsonian

SAMUEL PIERPONT LANGLEY

SAMUEL PIERPONT LANGLEY (1834–1906) rose to prominence as an astrophysicist working at the Allegheny Observatory in Pennsylvania. Recognized as one of America's leading scientists, he was appointed to the prestigious position of Secretary of the Smithsonian Institution in Washington, D.C., in 1887.

Langley began investigating the practicality of flight in the 1880s and continued his experiments at the Smithsonian, exploiting its resources. He progressed from building small models powered by rubber bands to larger steam-powered "Aerodrome" models.

In stark contrast to the Wright brothers, Langley developed no hands-on experience of either building flying machines or piloting them. His manned airplane, the Great Aerodrome, was the product of money and bureaucratic organization applied to the problem of flight. Its very public failure in 1903 was a crushing blow to the vanity of a proud man.



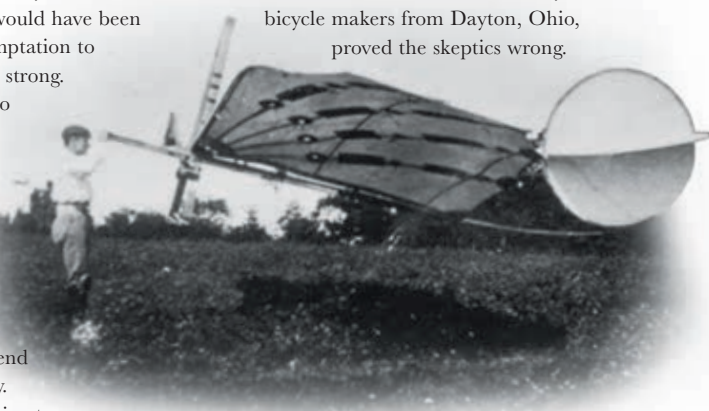
MAN OF LETTERS

A respected public figure with an impressive reputation as a scientist, Langley was humiliated by the Aerodrome fiasco.

workshops. The project ended up way over budget and four years behind schedule. And the huge flying machine that resulted simply did not work. Aerodynamically and structurally unsound, with no adequate control system, it twice plunged straight from the launch into the Potomac, taking Langley's reputation with it (see left).

The failure of this government-funded project conducted by America's leading scientist caused many people to conclude that heavier-than-air flight would never be a reality. Ironically, a mere nine days after Langley's last failed attempt with the Great Aerodrome in December 1903, two

bicycle makers from Dayton, Ohio, proved the skeptics wrong.



BRITISH EXPERIMENTER

British pioneer Percy Pilcher demonstrates his Bat glider, which he developed under the influence of Lilienthal. At the time of his death in 1899, he was experimenting with putting an engine on a glider – the path to powered flight that the Wrights would follow.